

Hardware, software and AI for STEM education — engineered end-to-end, under one roof.

Curio Makerlabs builds a complete learning ecosystem: an industrial-grade controller, a modular PCB construction system, a plug-and-play sensor library, an in-house coding studio, and AI tools with which students train and deploy real machine-learning models. One platform carries a learner from a first circuit to a working, real-world product.

01 OUR OWN HARDWARE

CPU, sensors and structural parts designed and manufactured by Curio — not re-badged imported kits.

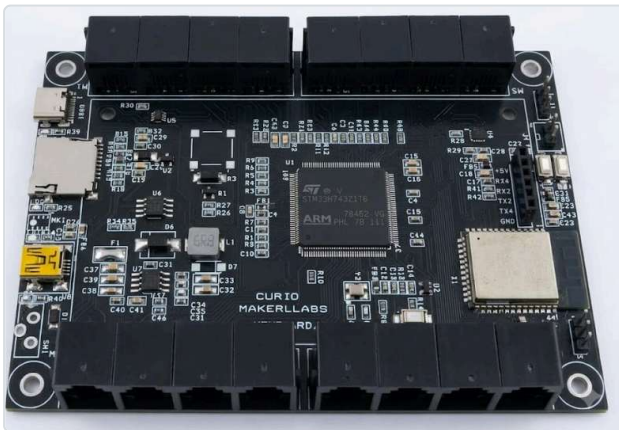
02 OUR OWN SOFTWARE

A complete coding studio developed in-house: blocks to text code, live data, wireless programming, teacher tools.

03 OUR OWN AI

Model training for image, voice, pose and gesture; an LLM copilot; and a RAG-based AI mentor for every student.

THE CURIO CPU



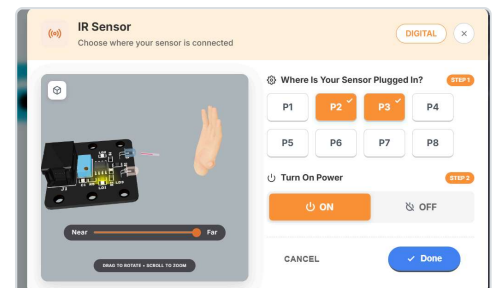
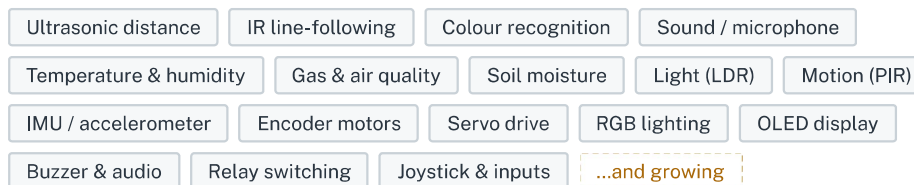
CURIO CPU — DESIGNED, ENGINEERED & ASSEMBLED IN-HOUSE

One controller, engineered for classrooms

PROCESSOR	Industrial-grade 32-bit ARM Cortex-M7 @ 480 MHz — real-time headroom for robotics, control loops and on-device AI workloads.
CONNECTIVITY	Onboard Wi-Fi + Bluetooth for wireless programming, live dashboards, app control and IoT projects.
PLUG & PLAY I/O	16 RJ sensor ports with polarised connectors — no jumper wires, no wiring errors, no short circuits in class.
EXPANSION	USB-C programming, microSD data logging, UART breakout and headers for advanced builds.
BUILT TO LAST	Protected onboard power, classroom-safe low voltage, rugged connectors rated for daily student use.

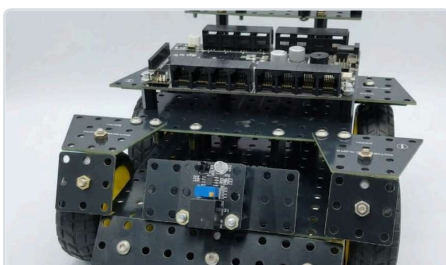
PLUG-AND-PLAY SENSOR ECOSYSTEM

Every module clicks into the same RJ ports — no soldering, no loose wires, no damaged pins. In our software, students pick the port on an interactive 3D view of each sensor, so wiring becomes a ten-second step.



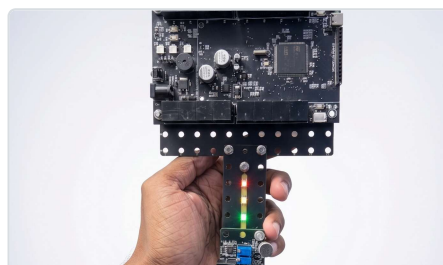
IN-SOFTWARE SETUP — PICK A PORT ON A LIVE 3D SENSOR

FROM KIT TO REAL-WORLD PRODUCT



Gesture-controlled smart car

Avoids obstacles with ultrasonic sensing and steers by hand gestures through a student-trained AI model — a complete real-time control loop.



Smart blind stick

An assistive-technology build that detects obstacles and warns with sound and light — students engineering for the visually impaired.

The chassis is a circuit board.

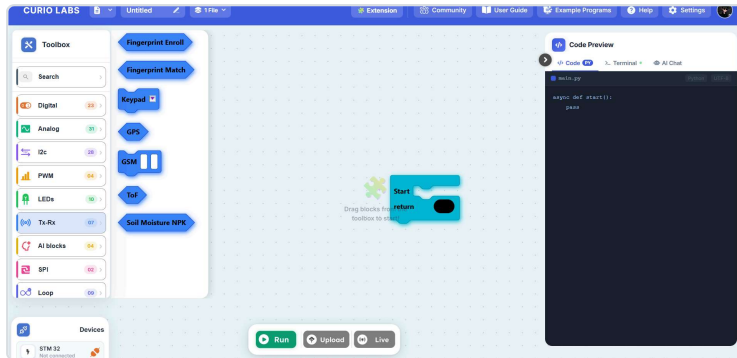
Plates, beams and brackets are precision PCB parts — students bolt together robots the way engineers prototype real products. Every build disassembles back into the kit, ready for the next idea.

One kit, unlimited possibilities

- Agriculture — soil, irrigation & weather monitors
- Environment — air-quality & pollution stations
- Smart city — traffic, lighting & parking models
- Health & accessibility — assistive devices
- Automation — voice & app-controlled systems

OUR SOFTWARE – BUILT ENTIRELY IN-HOUSE

Every screen of our platform is written by our own engineers, so it evolves with our hardware, our curriculum and feedback from real classrooms — no third-party licences, no lock-in.



BLOCK PROGRAMMING WITH LIVE CODE PREVIEW, TERMINAL & AI CHAT

Blocks → real code. Drag-and-drop coding with a live code preview — students graduate to text programming in the same window.

Run · Upload · Live. One-click flashing over Wi-Fi or USB and a live mode for instant feedback — no driver setup on school computers.

A deep toolbox. Digital, analog, I²C, PWM, displays, GPS, GSM, fingerprint, soil NPK, AI blocks and more — every module has its block.

Terminal & live data. A real terminal and streaming sensor readings — students see the data behind every project.

Examples, guides & community. Ready programs, a full user guide and a student community, one click away.

Classroom-friendly. Lightweight and low-bandwidth — built for ordinary school computers, not high-end labs.

THE AI LAB – STUDENTS TRAIN REAL MODELS

A Teachable-Machine-style workflow built into our software: collect samples, train a model in minutes, and deploy it straight to the build. Students don't just use AI — they make it, and watch it drive motors, lights and robots in real time.

Image

Camera classification — waste sorting, quality checks, smart door systems.

Voice

Speech commands & sound recognition — voice-controlled robots and alerts.

Pose

Full-body pose detection — posture alerts, fitness trainers, interactive games.

Gesture

Hand-gesture models that steer cars, arms and lighting — touch-free control.

LLM INTEGRATION

A coding copilot inside the studio

Students describe an idea in plain language — “*sound the alarm when gas crosses a limit*” — and the integrated LLM drafts the program, explains every step, and helps debug errors in words a learner understands. Ideas become working code faster, and the explanation teaches as it goes.

RAG-BASED AI MENTOR

Curio Mentor — a private tutor for every student

A retrieval-augmented AI grounded in our own curriculum, project guides and kit documentation — so answers are accurate, age-appropriate and syllabus-aligned. It guides with hints and next steps rather than ready-made answers, is available round the clock, and lets one instructor support an entire lab.

WHAT A CURIO DEPLOYMENT INCLUDES

Kits & spares

Classroom sets with spare-part support

Curriculum

Graded coursework mapped to class levels

Teacher training

Onboarding & certification for instructors

Software & AI

Coding studio, AI Lab & Mentor for every seat

Support

On-ground and remote assistance

Outcomes

Progress reporting for administrators

WHY CURIO FOR PUBLIC EDUCATION

NEP 2020 ALIGNED

Experiential, hands-on learning — coding, AI and 21st-century skills from middle school onward.

ATL & LAB READY

A drop-in platform for Atal Tinkering Labs, robotics labs and skill centres — one vendor, the complete stack.

MAKE IN INDIA

Hardware designed and manufactured indigenously; software and AI developed in-house — self-reliant supply, fast repair.

BUILT TO SCALE

Standardised kits, teacher certification and remotely manageable software make district-wide rollouts practical.

We would welcome the opportunity to demonstrate the platform in person — pilot deployments can be running in selected schools within weeks.